



# Configuring Secure Remote Desktop Connections via Fortinet VPN

Krish Thapaliya

## **Purpose**

This guide/analysis was created to introduce the configuration and deployment of a Fortinet Virtual Private Network (VPN) to enable secure remote resource sharing. This guide explains how to connect a remote client on an external private network to a host workstation residing behind a FortiGate firewall. By setting up this network architecture, users will understand how to establish a secure encrypted tunnel across different networks, allowing for Remote Desktop Protocol sessions between Windows devices. Doing these deployment steps provides essential on the site experience for network administrators tasked with managing secure remote workforces and site to site resource access.

## **Background**

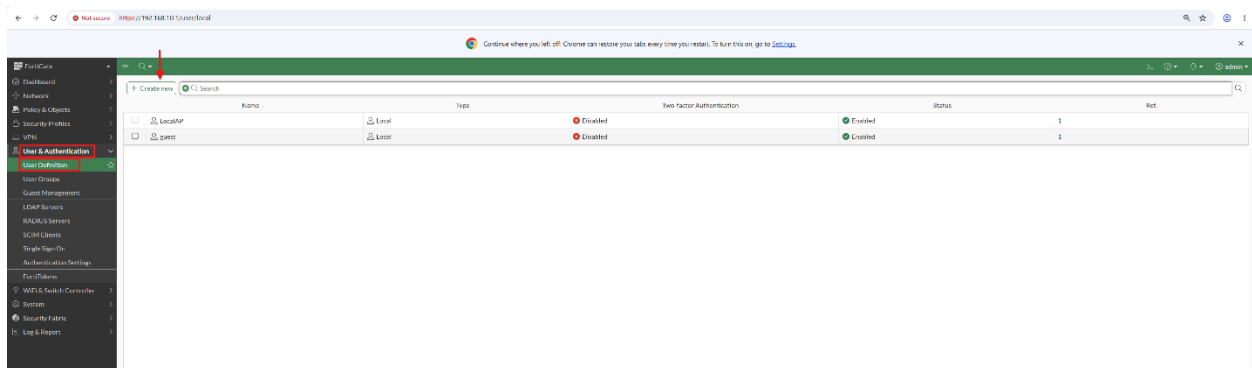
There are 2 main technologies and protocols that this guide/analysis will be explaining on how to both configure and utilize in a real applicable network. These 2 technologies are an IPsec VPN tunnel and the Remote Desktop Protocol (RDP). They are both categorized as Layer 7 application and session layer protocols. These technologies distinguish themselves by targeting specific layers in the OSI model to manipulate and route network traffic across different boundaries by exploiting security rules, user authentication, and encryption standards. These technologies happen across different private network segments, allowing remote devices to securely pass data or bypass physical geographic restrictions set, without needing to be on the same local physical network.

The first technology that is explained in this guide is the Fortinet IPsec VPN tunnel. This is a secure connection that is conducted on a computer network, where a remote user on an external network connects back to a protected internal network. When any new device is outside of an office network, the device itself cannot communicate with local resources for identification and security purposes but with the VPN tunnel, the FortiGate firewall will be able to automatically assign a virtual IP address to the remote device from a pre configured VPN pool, essentially granting the remote device access to the network as if it were physically plugged in. The reason that this connection is able to work on a network is because this setup uses the fact that a dedicated client, known as FortiClient, can establish a secure connection with the FortiGate firewall. The VPN connection itself uses IPsec protocols and authentication headers to trick the external network routing into believing

that the remote computer is part of the local office network, causing the firewall to securely pass encrypted traffic back and forth.

The second technology that is explained in this guide is the Remote Desktop Protocol (RDP). This is an application where a user sends GUI inputs onto a remote host machine. This will make the network associate the local user's mouse and keyboard inputs with the target computer making any display traffic meant for that host to be sent to the remote user instead. RDP also allows a user to fully manipulate all the files and applications on the remote host and is used as a standard administrative tool by network administrators. Remote Desktop normally functions by first having a network device sending an initial connection packet throughout the network. The second network device with the IP address listed in the request will then respond with an encrypted session reply that contains the display interface associated with its operating system. This configuration works because the FortiGate firewall keeps a policy table that stores individual security rules on the network to allow traffic to cross from the incoming IPsec VPN tunnel interface to the internal local area network, ensuring that only users with authorized credentials can gain access.

## Lab Commands



1. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## Users/Groups Creation Wizard

1 User Type > 2 Login Credentials > 3 Contact Info > 4 Extra Info

- Local User
- Remote RADIUS User
- Remote TACACS+ User
- Remote LDAP User
- FSSO
- FortiNAC User
- SAML User

< Back

Next

Cancel

2. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## Users/Groups Creation Wizard

✓ User Type → 2 Login Credentials → 3 Contact Info → 4 Extra Info

Username

Password  

 Online Guides

 Relevant Docum

 Video Tutorials

 Fortinet Commu

 Join the Discussi

< Back

Next

Cancel

3. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## Users/Groups Creation Wizard

✓ User Type > ✓ Login Credentials > **3 Contact Info** > 4 Extra Info

☐ Two-factor Authentication



< Back

Next

Cancel



4. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## Users/Groups Creation Wizard

✓ User Type > ✓ Login Credentials > ✓ Contact Info > **4 Extra Info**

User Account Status **Enabled** Disabled

User Group ☒

### Select Entries

Search  **+ Create**

Guest-group  
Mr.Beast

Close

< Back

Submit

Cancel

5. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

Users/Gr New User Group



User A

User G

Name **VPNGroup**

Type **Firewall**  
Fortinet Single Sign-On (FSSO)  
RADIUS Single Sign-On (RSSO)  
Guest  
scim

Members +

Remote Groups

| <div><div>+ Add</div><div>Edit</div><div>Delete</div></div> |            |
|-------------------------------------------------------------|------------|
| Remote Server                                               | Group Name |
| No results                                                  |            |
| 0                                                           |            |



OK Cancel

6. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## Users/Groups Creation Wizard

✓ User Type > ✓ Login Credentials > ✓ Contact Info > 4 Extra Info

User Account Status ↑ Enabled ↓ Disabled

User Group ✓

VPNGroup ✕

Select Entries ✕

Search + Create

Guest-group

Mr.Beast

VPNGroup ✎

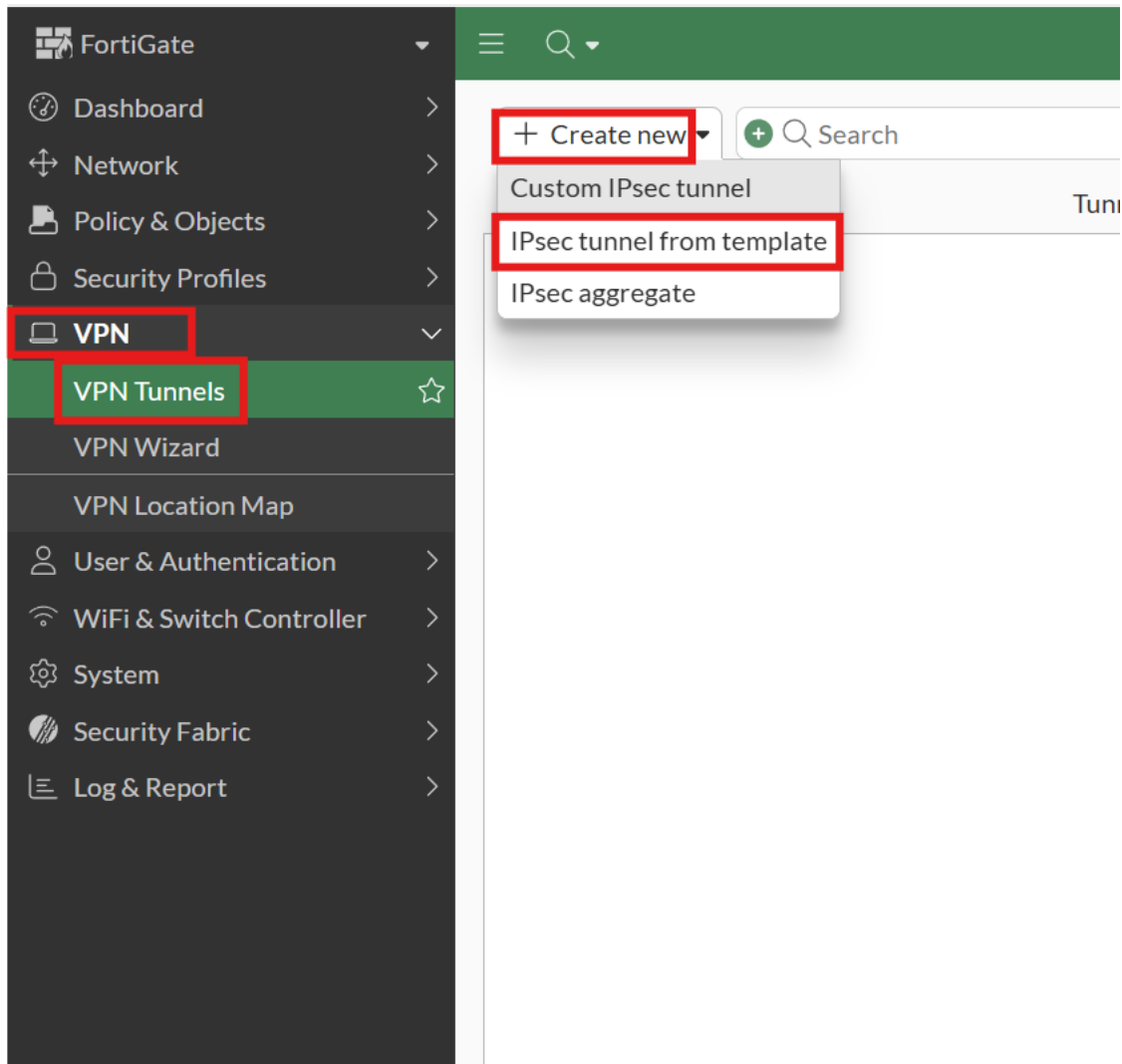
Close

< Back

Submit

Cancel

7. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above



8. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## VPN Wizard

The IPsec VPN tunnel wizard gets you started with templates for common use cases. If the templates below do not fulfill your needs, fully custom tunnels can be set up. For convenience, it is also possible to start with a template and convert it to a fully custom tunnel after.

Tunnel name

VPNRemoteRDP

Select a template ☐



Remote Site



VPN Tunnel



Local FortiGate

### Site to Site

Offices in multiple, fixed locations can establish secure connections with each other. Branch offices can also access the main office's intranet.

Supported remote peers:  

☐



Spokes



VPN Tunnels



Hub



### Hub and Spoke with ADVPN

VPN connections radiate from a central FortiGate unit (the hub) to multiple remote peers (the spokes). Traffic can pass between private networks that are each behind a hub or remote peer (spoke). Additionally, spoke to spoke traffic is possible via ADVPN tunnels.

Supported remote peers: 

☒



Remote  
Endpoints



VPN Tunnels



Local FortiGate

### Remote Access

Employees who need to access their company's network from off-site locations or users who want to securely connect to a private network from a public area frequently use this type of VPN.

Supported remote clients:    

☐



Begin



9. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## VPN Wizard - VPNRemoteRDP



Local FortiGate

### VPN tunnel

VPN client type **FortiClient** Native

Authentication method **Pre-shared key** Signature

Pre-shared key **CiscoLab**

IKE **Version 2** Version 1

Transport **UDP** Auto

NAT traversal **Enable** Disable Forced

Keepalive frequency **10**

EAP peer identification IKEv2 IDi payload **EAP identity request**

User authentication method **Phase 1 interface** Inherit from policy

**VPNGroup**

DNS Server **Use System DNS** Specify

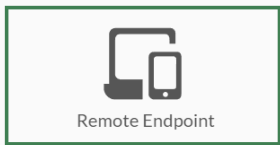


Next

Cancel

10. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## VPN Wizard - VPNRemoteRDP



Remote Endpoint



VPN Tunnel



Incoming



Local FortiGate



Internal

### VPN tunnel

VPN client type: **FortiClient** Native

Authentication method: **Pre-shared key** Signature

Pre-shared key: CiscoLab

IKE: **Version 2** Version 1

Transport: **UDP** Auto

NAT traversal: **Enable** Disable Forced

Keepalive frequency: 10

EAP peer identification: IKEv2 IDi payload **EAP identity request**

User authentication method: **Phase 1 interface** Inherit from policy

VPN Group: VPNGroup

DNS Server: **Use System DNS** Specify

### Remote endpoint

Addresses to assign to connected endpoints: 10.67.67.0-10.67.67.255

Subnet for connected endpoints: 255.255.255.0

### FortiClient settings

Security posture gateway matching: ☐

EMS SN verification: ☐

Save password: ☒

Auto Connect: ☐

Always up (keep alive): ☐



Next

Cancel

11. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above



+ VPN tunnel

+ Remote endpoint

Local FortiGate

Incoming interface that binds to tunnel Internet (wan)

Create and add interface to zone ☒

Local interface Office LAN (lan)

Local Address lan



Next Cancel

12. Follow the steps in the box and then do the final step pointed by the arrow as highlighted in the image above

## **Problems**

I found a couple of small configurations and system hurdles that prevented successful connectivity. The main problem I had, after creating the VPN tunnel, was at first unable to create the RDP session to the internal computer. After checking the target Windows machine, I realized that Remote Desktop was disabled by default in the system settings and that the Windows Defender Firewall was actively blocking incoming traffic on RDP port 3389. I fixed this by enabling Remote Desktop on the host device and modifying the local Windows firewall rules to allow inbound connections.

## **Conclusion**

In conclusion, this analysis/guide gave valuable experience in configuring a Fortinet VPN solution to facilitate secure, cross network workstation access. I now have a solid understanding of how a FortiGate firewall authenticates remote clients and bridges separate private networks together. Also, managing the interaction between network level security and host level Windows configuration showed the importance of checking both firewall policies and local OS settings.